

Beginner's Strategy for Simulating Bacterial Biomarkers in Gaussian and Obtaining Raman Spectra

Prepared for: SERS-Based Sepsis Biomarker Detection Project

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Molecules: N-acetylmuramic acid (MurNAc), Lipoteichoic Acid (LTA), Lipopolysaccharide (LPS)

Executive Summary

This step-by-step guide is designed for beginners entering computational chemistry and Raman spectroscopy simulation. It covers the complete workflow: molecular structure preparation → geometry optimization → vibrational frequency calculation → Raman spectrum generation and analysis using **Gaussian 16/09** software combined with **GaussView** (visualization) and **GaussSum** (spectral analysis).

Target Outcome: Generation of theoretical Raman spectra peaks (wavenumbers and intensities) that can be compared with experimental SERS data for biomarker identification and sepsis diagnostics.

Part 1: Pre-Simulation Setup & Prerequisites

1.1 Software Installation & Environment

Required Software:

- **Gaussian 16 or 09** – Quantum chemistry calculation engine
- **GaussView 6** or compatible – Molecular visualization and input file generation
- **GaussSum 3.0** – Post-processing and spectrum generation from Gaussian output
- **Avogadro 1.2+** (Free, Open-Source) – Alternative for initial structure drawing and format conversion
- **Text editor** (e.g., Notepad++, nano) – For manual input file editing
- **Spreadsheet software** (Excel, LibreOffice Calc) – For data visualization and peak analysis

Installation Tips:

- Install Gaussian and GaussView on a workstation or cluster with adequate RAM (minimum 8GB; 16GB+ recommended for larger molecules).
- GaussSum and Avogadro are free and platform-independent (Windows, Mac, Linux).
- Verify Gaussian installation by testing a simple water (H₂O) optimization job.

1.2 Computational Resources & Time Estimates

Task	Method	Basis Set	Time (Single Core)	Recommended Basis
Geometry Optimization	B3LYP-DFT	6-31G(d)	5–30 min (small), 2–4 hours (biomarkers)	6-31G(d) or 6-311G(d)
Frequency Calculation	B3LYP-DFT	6-31G(d)	10–60 min (small), 4–8 hours (biomarkers)	Same as opt
Single-Point Raman	B3LYP-DFT	6-311G(d)	15–120 min	For refined spectra

Estimated project timeline per biomarker: 12–24 hours wall-clock time on a modern multi-core workstation.

Part 2: Molecular Structure Preparation (Step 1–3)

Step 1: Obtain or Draw Initial Molecular Structure

For Beginners – Three Approaches:

Option A: Download Pre-Optimized Structures from PubChem

1. Visit <https://pubchem.ncbi.nlm.nih.gov>
2. Search for the molecule (e.g., "N-acetylmuramic acid" or use CID: 82756)
3. Download as **SDF** or **MOL** file format
4. Open in GaussView or Avogadro

PubChem CIDs for Target Molecules:

- N-Acetylmuramic acid: CID 82756 or 5462244
- Lipoteichoic acid (representative): CID 137349712
- Lipid A: CID 9877306
- Note: O-antigen has species-specific variability; use E. coli O111 or representative unit structures.

Option B: Manual Structure Drawing in Avogadro (Recommended for Learning)

1. Open **Avogadro** → File → New
2. Select **Draw Tool** (pencil icon)
3. Left-click to add atoms; select element from dropdown (C, H, O, N, P, S)
4. Click and drag to create bonds (adjust bond order: 1=single, 2=double, 3=triple)
5. Enable "Adjust Hydrogens" checkbox to auto-populate H atoms
6. Click **Extensions** → **Optimize Geometry** (uses UFF force field for rough pre-optimization)
7. Save as **.mol** file

Option C: Use GaussView to Build Structure

1. Launch **GaussView**
2. Select **Element Fragment** from toolbar
3. Click in the workspace to add atoms
4. Use **R-Group Fragment** to add substituents
5. Click **Edit** → **Clean Geometry** to regularize structure
6. Click **Edit** → **Symmetrize** to identify point group symmetry
7. Save as **.gjf** (Gaussian input file) or **.com**

For Complex Biomarkers (LTA, LPS): Start with the **smallest repeating unit** or **monomeric substructure** to reduce computational cost. For example:

- For LTA: model a **glycerol-phosphate dimer** or **single repeat unit** rather than the full polymer.
- For LPS: use **Lipid A core** alone or **Lipid A + one Kdo residue** as a tractable model.
- For MurNAc: use the **single residue** or **MurNAc-GlcNAc dipeptide**.

Step 2: Import Structure into GaussView and Verify Geometry

1. **Open GaussView**
2. **File** → **Open** → Select your **.mol**, **.sdf**, or **.gjf** file
3. **Verify initial structure:**
 - Inspect bond lengths (right-click on bond to measure)
 - Check for missing atoms or incorrect connectivity
 - Ensure stereochemistry is correct (wedge/dash bonds for stereocenters)
4. **Clean and Prepare:**
 - Click **Edit** → **Clean Geometry** (optimizes bond angles/lengths using force field)
 - Click **Edit** → **Symmetrize** (identifies point group symmetry; optional but informative)
5. **Save as GaussView File:**
 - **File** → **Save as .gv** (retains 3D structure for later use)

Step 3: Add Charge and Multiplicity

1. **In GaussView:**
 - Click **Molecule** → **Properties**
 - Set **Charge:** 0 (for neutral biomarkers; adjust if ionized)
 - Set **Multiplicity:** 1 (for closed-shell; use 2 for radicals)
 - Default: Most bacterial biomarkers at physiological pH are **charge 0–2– (anionic due to phosphates)**

2. For phosphate-rich biomarkers (LTA, LPS, MurNAc with phosphate):

- Physiological state: **Charge = -1 to -3** (deprotonated phosphates)
- Best practice: Model at both **neutral and anionic states** to compare SERS activity
- Consider **solvent effects** (aqueous environment) — set PCM (Polarizable Continuum Model) in Gaussian input

Part 3: Geometry Optimization (Step 4–6)

Step 4: Generate Gaussian Input File for Geometry Optimization

In GaussView:

1. Click **Calculate** → **Gaussian Calculation Setup**

2. A new window appears: **Gaussian Calculation Setup**

3. **Job Type Tab:**

- Select **Opt** (Geometry Optimization)
- *Do NOT select Freq yet; only do Opt first*

4. **Method Tab:**

- **Functional:** B3LYP (hybrid DFT; good balance of speed and accuracy)
- **Basis Set:** 6-31G(d) (good starting point; 6-311G(d) for refined)
- **Solvent Model (optional):** PCM=UFF (for aqueous environment like sepsis)

5. **Additional Tab (Optional):**

- Check **Symmetry:** Auto (lets Gaussian detect symmetry)
- Check **Unit cell** (if modeling periodic systems; not applicable for isolated molecules)

6. **Route Line Preview:** Should show:

```
# B3LYP/6-31G(d) Opt
```

7. Click **Submit** (or save input file first)

Alternative: Manual Input File Creation

Create a text file with **.com** or **.gjf** extension:

```
%chk=MurNAc_opt.chk
%mem=8GB
%nproc=4
# B3LYP/6-31G(d) Opt

N-acetylmuramic acid geometry optimization

0 1
C      0.00000      0.00000      0.00000
```

```
0      1.20000      0.00000      0.00000
... (Cartesian coordinates of all atoms)
```

Key Input File Components:

- `%chk=filename.chk` – Checkpoint file (stores intermediate data)
- `%mem=8GB` – Memory allocation (adjust based on system)
- `%nproc=4` – Number of processors
- `# B3LYP/6-31G(d) Opt` – Route line (method, basis, job type)
- **Title line** (any descriptive text)
- **Charge and Multiplicity** (e.g., `0 1` for neutral singlet)
- **Molecular geometry** (Cartesian or Z-matrix coordinates)

Step 5: Submit Optimization Job to Gaussian

On a Local Machine:

1. In GaussView, click **Submit** (after filling in the Calculation Setup)
2. Alternatively, from terminal:

```
g16 MurNAc_opt.com MurNAc_opt.log
```

On a High-Performance Computing (HPC) Cluster:

1. Prepare a job submission script (e.g., SLURM, PBS):

```
#!/bin/bash
#SBATCH --job-name=MurNAc_opt
#SBATCH --nodes=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=16G
#SBATCH --time=12:00:00

module load gaussian
g16 MurNAc_opt.com MurNAc_opt.log
```

2. Submit:

```
sbatch submit_job.sh
```

Monitor Progress:

- Check logfile for keywords: "Optimization completed" (success) or "Error" (failure)
- GaussView can display geometry update in real-time
- Typical warnings: "Initial guess from Harris functional" (normal)

Step 6: Verify Optimization Convergence

1. Open output file (.log or .out) in GaussView:

- File → Open → Select **.log** file
- GaussView displays optimized structure

2. Check Key Convergence Indicators:

- **No negative frequencies:** Confirms structure is at a minimum (not a saddle point)
- **SCF Convergence:** Last line should state "convergence achieved"
- **Energy:** Final energy value shown in Hartrees (e.g., -123.456789)

3. Common Issues:

- **Imaginary frequencies during optimization:** Molecule geometry is not at a minimum; re-optimize with tighter convergence criteria
- **No convergence after 100 steps:** Increase max steps with `Opt=(MaxCycles=200)` in route line

Part 4: Vibrational Frequency & Raman Calculation (Step 7–8)

Step 7: Generate Input File for Freq=Raman Calculation

Critical: Always run frequency calculations using the **same method and basis set** as the optimization. Do NOT compute frequencies at a different level of theory than the optimized structure.

In GaussView (Recommended Workflow):

1. File → Open the optimized structure (**MurNAc_opt.log** from Step 5)

2. Click **Calculate** → **Gaussian Calculation Setup**

3. Job Type Tab:

- Select **Freq** (Frequency calculation)
- Check **Raman** (to calculate Raman activity; NOT just IR)

4. Method Tab:

- **Functional:** B3LYP (same as optimization)
- **Basis Set:** 6-31G(d) (same as optimization)

5. Route Line Preview:

```
# B3LYP/6-31G(d) Freq=Raman
```

6. Use Checkpoint: Enable **geom=allcheck** to read optimized geometry from .chk file

- Route line becomes:

```
# B3LYP/6-31G(d) geom=allcheck guess=read Freq=Raman
```

7. Click **Submit**

Manual Input File (Using Checkpoint):

```
%oldchk=MurNAc_opt.chk
%chk=MurNAc_freq.chk
%mem=8GB
%nproc=4
# B3LYP/6-31G(d) geom=allcheck guess=read Freq=Raman

Frequency and Raman calculation for MurNAc

0 1
```

(Leave molecular coordinates blank; Gaussian reads from .chk file)

Step 8: Execute Frequency/Raman Calculation

Submit job (same as Step 5):

```
g16 MurNAc_freq.com MurNAc_freq.log
```

Expected Output (from .log file):

- Vibrational frequencies (cm^{-1}) for each normal mode
- Raman activity (in $\text{\AA}$) for each mode
- IR intensities (km/mol)
- Force constants

Part 5: Spectrum Generation & Analysis (Step 9–10)

Step 9: Extract Raman Spectrum Using GaussSum

GaussSum is the key tool for converting raw vibrational data into publication-ready spectra.

Step-by-Step in GaussSum:

1. **Launch GaussSum** (free, open-source; www.gausssum.sourceforge.net)
2. **File** → **Open** → Select **MurNAc_freq.log** (the Gaussian frequency output)
3. **Left Panel** → **Select "Frequencies"**
4. **Set Raman Spectral Parameters:**
 - **Start wavenumber:** 0 cm^{-1}
 - **End wavenumber:** 3500 cm^{-1} (covers most vibrational modes)
 - **Number of points:** 1000 (higher = finer resolution)

- **FWHM (Full Width Half Maximum):** 10 cm^{-1} (typical for Raman)
- **Wavelength (for Raman intensity correction):**
 - 785 nm (common in SERS; NIR laser)
 - 532 nm (visible laser)
 - 1064 nm (Nd:YAG)
- **Temperature:** 298 K (or match your experimental conditions)
- **Scaling factor:** 0.98 (empirical correction for DFT overestimation; consult literature)

5. Click "GaussSum icon" → generates Raman spectrum

6. Output:

- Automatic plot displayed
- Data exported to **IRSpectrum.txt** and **RamanSpectrum.txt** (in gausssum3 subfolder)

Key GaussSum Outputs:

- **RamanSpectrum.txt:** Contains wavenumber (cm^{-1}), Raman activity, intensity
- **Format:** Tab-separated; easily imported into Excel or Origin for visualization

Step 10: Peak Identification & Data Analysis

In Excel or LibreOffice:

1. Import RamanSpectrum.txt:

- File → Open → Select RamanSpectrum.txt
- Select "Tab-delimited" import option

2. Create Peak Table:

Wavenumber (cm^{-1})	Raman Intensity	Functional Group Assignment	Biomarker Relevance
580	High	P–O stretch (phosphate)	LTA, LPS (diagnostic)
1050	Medium	C–O stretch (glycosidic)	MurNAc, LTA, LPS
1200	Medium	Amide III (peptide)	MurNAc
1600	High	Amide II (N–H bend)	MurNAc
2900	Medium	C–H stretch	All molecules

3. Plot Raman Spectrum:

- X-axis: Wavenumber (cm^{-1})
- Y-axis: Raman Intensity (or Activity)
- Mark prominent peaks with functional group labels
- Add experimental reference spectra (if available) for comparison

4. Quantitative Analysis:

- **Peak position consistency:** Compare calculated vs. experimental $\pm 5\text{--}10 \text{ cm}^{-1}$

- **Intensity ratios:** Phosphate peaks should dominate for LTA/LPS
- **Frequency scaling:** If discrepancy >10%, apply correction factor (e.g., $0.98 \times \text{frequency}$)

Part 6: Troubleshooting & Best Practices

Common Issues & Solutions

Issue	Cause	Solution
Job terminates: "Optimization did not converge"	Geometry too far from minimum	Increase MaxCycles=300; try finer geometry input
Imaginary frequency appears	Saddle point, not minimum	Re-run optimization; check molecular geometry for errors
No Raman peaks generated	Missing Freq=Raman keyword	Ensure route line includes Freq=Raman, not just Freq
Huge discrepancy with experimental data	Wrong scaling factor; poor basis set	Apply 0.98 scaling; try 6-311G(d) basis; check if molecule needs solvent model
GaussSum crashes or no output files	Corrupted .log file	Re-run Gaussian calculation; check file completeness
Memory errors in Gaussian	Job exceeds available RAM	Reduce %mem or split into smaller sub-molecules

Best Practices for Accurate Results

1. **Always use the same method for both Opt and Freq jobs** (consistency is critical)
2. **For biomarkers with phosphates:** Include solvent model (PCM for aqueous)
3. **Validate geometry:** Inspect bond lengths/angles against literature values
4. **Scale frequencies:** Apply empirical scaling (typically 0.95–0.98 for B3LYP/6-31G)
5. **Experimental comparison:** If experimental Raman is available, refine basis set or functional
6. **Document all parameters:** Keep record of method, basis, temperature, laser wavelength for reproducibility

Part 7: Advanced Considerations for Biomarker Molecules

Handling Complex & Large Biomarkers

Problem: Full LPS or LTA polymers are computationally intractable (hundreds of atoms, hours-to-days runtime).

Solution Strategy:

1. **Model Only the Diagnostic Unit:**
 - **MurNAc:** Use single residue ✓ Tractable

- **LTA:** Use **glycerol-phosphate monomer or dimer** (not full polymer) ✓ ~10–30 min
- **LPS:** Use **Lipid A core** (not full O-antigen) ✓ ~2–4 hours
- **O-antigen:** Use **representative disaccharide repeat unit** ✓ ~30–60 min

2. Validate Approach:

- Compare predicted SERS peaks of truncated model with full biomarker spectra (if literature available)
- Expect diagnostic peaks (phosphate at $\sim 580\text{ cm}^{-1}$, amide bands) to remain representative

3. Consider Multiple Conformations:

- Biomarkers may have multiple low-energy conformers
- Run optimization from 2–3 different starting geometries
- Compare Raman spectra across conformers
- Report most stable (lowest energy) conformation

Ionization States & pH Effects

Bacterial biomarkers at physiological pH (7.4) are typically **deprotonated and anionic**:

- Phosphate groups ($-\text{PO}_4^{2-}$ vs. $-\text{PO}_4^{3-}$)
- Carboxylate groups ($-\text{COO}^-$)

Recommendation: Model both protonated and deprotonated forms:

- **Neutral form:** Charge = 0 (simpler, faster)
- **Anionic form:** Charge = -1 , -2 , or -3 (more realistic for sepsis biomarkers)
- Compare Raman shifts; phosphate peaks are most diagnostic and sensitive to protonation

Molecular Solvent Effects (Optional but Recommended)

For physiologically relevant predictions, include aqueous solvation:

In Gaussian input route line, add:

```
# B3LYP/6-31G(d) Opt SCRF=(PCM,Solvent=Water)
```

Effect: Slight red-shift of frequencies; improves agreement with experimental aqueous SERS data.

Part 8: Integration with SERS Experimental Campaign

Comparing Theory to Experiment

1. **Collect experimental SERS spectra** of reference biomarkers (MurNAc, LTA, Lipid A) using same laser wavelength as simulated in GaussSum (e.g., 785 nm)
2. **Align peaks:** Shift calculated spectrum if systematic offset observed (e.g., 1–2% frequency shift common)
3. **Build spectral library:** Create database of theoretical + experimental peaks for each biomarker
4. **Clinical validation:** Use combined spectral library to identify unknowns in patient samples

Multiplexing Strategy

For sepsis diagnostics, you may target multiple biomarkers simultaneously:

Biomarker	Diagnostic Peak (cm ⁻¹)	Gram-Positive/-Negative	SERS Enhancement
MurNAc	1200 (amide III)	Gram-Positive	Medium
LTA	580 (P–O)	Gram-Positive	Very High
Lipid A	580, 1200	Gram-Negative	Very High
O-antigen	1050–1100 (C–O)	Gram-Negative	Medium

Part 9: Quick Reference Workflows

Minimal Workflow (Learning; ~1–2 hours total)

1. Download N-Acetylmuramic acid (CID 82756) from PubChem as SDF
2. Open in GaussView → Geometry Optimization B3LYP/6-31G(d) (~30 min)
3. Frequency/Raman calculation on optimized structure (~30 min)
4. Open output in GaussSum → Generate Raman spectrum (2 min)
5. Plot and compare to literature

Full Workflow (All three biomarkers; ~24–48 hours total)

For each biomarker (MurNAc, LTA, Lipid A):

1. Prepare structure (30 min)
2. Geometry optimization (2–4 hours)
3. Frequency/Raman (4–8 hours)
4. Spectrum generation & analysis (30 min)
5. Comparative analysis across all three (1 hour)

Part 10: References & Further Resources

Key Literature

1. **Gaussian Manual:** <https://gaussian.com/g16main/>
2. **GaussView Manual:** <https://gaussian.com/gaussview6/>
3. **GaussSum Documentation:** <https://gausssum.sourceforge.net/>
4. **Vibrational Analysis:** Ochterski, J. W. (1999). *Vibrational Analysis in Gaussian* [comprehensive reference on frequency calculations]
5. **Raman Spectroscopy Guide:** Krishnakumar, V., et al. (2004). Determination of Raman intensity using different methods. *J. Mol. Struct.* 702, 9–21.

Online Tutorials & Community

- **YouTube:** Search "Gaussian Raman spectrum tutorial" or "GaussSum workflow"
- **Forum:** Gaussian Computational Chemistry Community (LinkedIn, ResearchGate)
- **PubChem:** <https://pubchem.ncbi.nlm.nih.gov> (structure and property database)

Software Downloads

- **Gaussian 16:** <https://gaussian.com/> (commercial; check university licensing)
- **GaussView 6:** Bundled with Gaussian
- **GaussSum:** <https://gausssum.sourceforge.net/> (free, open-source)
- **Avogadro:** <https://avogadro.cc> (free, open-source; excellent for learning)

Appendix: Example Input Files

Example 1: MurNAc Geometry Optimization Input

```
%chk=MurNAc_opt.chk
%mem=8GB
%nproc=4
# B3LYP/6-31G(d) Opt

N-acetylmuramic acid optimization

0 1
C   -0.5211  -0.2198   0.0149
C    0.9854  -0.2207   0.0088
O    1.5471   0.9833  -0.0356
O    1.6831  -1.2701   0.0418
N   -1.1893   0.9920  -0.0359
C   -2.6168   1.0654  -0.0452
O   -3.3192   0.0608  -0.0141
C   -3.2286   2.3840  -0.0864
H   -0.5850  -0.7283   0.9552
```

```
... (additional atoms)
```

Example 2: Frequency/Raman Input (Using Checkpoint)

```
%oldchk=MurNAc_opt.chk  
%chk=MurNAc_freq.chk  
%mem=8GB  
%nproc=4  
# B3LYP/6-31G(d) geom=allcheck guess=read Freq=Raman  
  
Frequency and Raman for MurNAc  
  
0 1
```

Final Checklist Before Submission

- ☐ Molecule structure verified and saved
- ☐ Charge and multiplicity set correctly
- ☐ Sufficient computational resources allocated
- ☐ Method and basis set consistent across Opt and Freq jobs
- ☐ GaussSum configured with correct laser wavelength
- ☐ Reference literature/experimental data ready for comparison
- ☐ Outputs backed up (especially .chk and .log files)

End of Guide

For further guidance or troubleshooting specific to your biomarker system, consult literature on DFT calculations for polysaccharides and glycolipids, or reach out to computational chemistry specialists.

[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [37] [38] [39] [40] [41] [42] [43] [44] [45] [46] [47] [48] [49] [50] [51] [52] [53] [54] [55] [56] [57] [58] [59] [60] [61] [62] [63] [64] [65] [66] [67] [68] [69] [70] [71] [72] [73] [74] [75] [76] [77] [78] [79] [80] [81] [82] [83] [84] [85] [86] [87] [88] [89] [90] [91] [92] [93] [94] [95]

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1. <https://link.springer.com/10.1007/s00339-023-07083-1>
2. <https://pubs.acs.org/doi/10.1021/acs.jctc.3c01343>
3. <https://xlink.rsc.org/?DOI=D3CP04960D>
4. <https://arxiv.org/pdf/2201.07586.pdf>
5. <https://arxiv.org/pdf/1909.03142.pdf>
6. <http://arxiv.org/pdf/2303.10228.pdf>
7. <https://arxiv.org/pdf/2409.06597.pdf>

8. <https://joss.theoj.org/papers/10.21105/joss.04164.pdf>
9. <http://arxiv.org/pdf/2308.04308.pdf>
10. <https://onlinelibrary.wiley.com/doi/10.1002/jcc.27027>
11. <https://www.youtube.com/watch?v=5-2J8306XI0>
12. <https://pubs.aip.org/jva/article/42/4/043405/3294350/Lattice-defects-distribution-of-H-implanted-4H-SiC>
13. <https://gaussian.com/g16main/>
14. https://dspace.mit.edu/bitstream/handle/1721.1/152352/5-68j-spring-2003/contents/study-materials/g94_tutorial_v2.pdf
15. <https://www.youtube.com/watch?v=TzjYqWsGMho>
16. <https://onlinelibrary.wiley.com/doi/10.1002/qua.70072?af=R>
17. <https://www.dornshuld.chemistry.msstate.edu/research/first-gaussian-job/>
18. <https://www.scm.com/doc/Tutorials/ExternalPrograms/QEIRamanSpectra.html>
19. <https://gaussian.com/relnotes/>
20. <https://gaussian.com/wp-content/uploads/dl/vib.pdf>
21. <https://gaussian.com/freq/>
22. <https://link.springer.com/10.1007/s11082-023-04877-8>
23. <https://www.ssrn.com/abstract=4445044>
24. <https://journaljenrr.com/index.php/JENRR/article/view/201>
25. [https://wseas.com/journals/msa/2023/a06msa-003\(2023\).pdf](https://wseas.com/journals/msa/2023/a06msa-003(2023).pdf)
26. <http://link.springer.com/10.1134/S1070428019020192>
27. http://www.ijnc.ir/article_32806.html
28. http://www.physchemres.org/article_110988.html
29. <https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/slct.202500161>
30. <https://link.springer.com/10.1007/s13538-023-01367-w>
31. <https://link.springer.com/10.1007/s00894-024-05866-8>
32. <https://www.tandfonline.com/doi/full/10.1080/10406638.2023.2195191>
33. <https://arxiv.org/pdf/2211.15299.pdf>
34. <https://onlinelibrary.wiley.com/doi/10.1002/qua.70072>
35. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7997400>
36. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7997400/>
37. <https://arxiv.org/pdf/2303.14280.pdf>
38. <https://arxiv.org/pdf/2105.00589.pdf>
39. <http://arxiv.org/pdf/2411.12566.pdf>
40. <https://joss.theoj.org/papers/10.21105/joss.05148.pdf>
41. <http://arxiv.org/pdf/2209.06069.pdf>
42. <https://www.uprm.edu/ccs-cicsa/wp-content/uploads/sites/183/2023/04/SOP-Creating-a-Molecule-GaussView.pdf>
43. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2176076/>
44. <https://hpc-wiki.info/hpc/Gaussian>

45. <https://link.springer.com/10.1007/s10853-022-07011-3>
46. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5319398/>
47. <https://www.youtube.com/watch?v=EJvUcfdnhpE>
48. <https://www.youtube.com/watch?v=dqKWfXsEcWs>
49. <https://pubs.acs.org/doi/10.1021/acs.jpcc.4c03491>
50. <https://link.springer.com/10.1134/S1062873824709905>
51. <https://xlink.rsc.org/?DOI=D5AN00452G>
52. <https://www.semanticscholar.org/paper/b44369e5acbb81ee0f5d9c3181a0aa9fde6195f2>
53. <https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/jrs.70006>
54. <https://opg.optica.org/abstract.cfm?URI=boe-13-12-6763>
55. <https://www.mdpi.com/2218-1989/15/1/48>
56. <https://link.springer.com/10.1007/s00604-024-06831-1>
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